# When Does a Holonic Claim Have Empirical Content?

## A Minimal Constitutive Framework, with Demarcation as a Consequence

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**Abstract**

Holism—that a system is best understood through its place in a nested whole—recurs across the sciences: Koestler's *holon*, von Bertalanffy's *general systems*, Friston's Markov-blanketed *things*. Its strong form asserts that every system is holonically organised, with the universe as the maximal holon. I ask the only version of the question that can be wrong: *when does calling a system a holon forbid an observation?* The answer is a **minimal constitutive framework**—the smallest set of epistemic commitments, fixed before the data, under which a holonic claim has empirical content at all. There are three, and they are inseparable: a declared **module partition**, a declared **timescale**, and a **contraction counterfactual** (Bousquet–Slotine's $\varepsilon < \varepsilon_c$). Pre-registration is their institutional implementation; Simon's near-decomposability supplies the measurable content. The framework is claimed as **necessary only**: whether such commitments *suffice* to block Putnam-triviality inherits an open problem (Putnam–Chalmers) the paper states rather than settles. **Demarcation follows as a consequence**—a system for which no admissible partition clears the signature gains no holonic content—and I fix the contraposition class it excludes (turbulence, self-organised criticality, quantum non-separability). The capstone—"the universe is the principal holon"—is shown to be a category error on the holon's own Janus definition. A worked **self-falsification** from a built holonic system is reported in the appendix.

# 1 Introduction

## 1.1 A thesis that keeps being rediscovered

The intuition is old and recurrent: a system is not fully intelligible in isolation, because it is simultaneously a *whole* (with its own integrity) and a *part* (of something larger), and its behaviour is shaped from both directions at once. Koestler (1969) gave the intuition its sharpest name—the **holon**, the Janus-faced entity that "display[s] both the autonomous properties of wholes and the dependent properties of parts"—and the **holarchy**, the nested order of holons. Von Bertalanffy's General Systems Theory (Bertalanffy 1968) had already made the more ambitious move of asserting that such organisational principles hold *across all domains*. Half a century later the same structure reappears, formally re-equipped, in Karl Friston's Free Energy Principle, which individuates "every 'thing' that can be distinguished from other things" by a **Markov blanket** and stacks those blankets "all the way down" (Friston 2019).

This paper takes seriously a maximally ambitious version of the intuition, of the kind a systems-minded thinker is naturally led to:

> *Every behaviour of any system can be explained as a holonic system and/or a complex dynamical system. Each system functions individually while simultaneously being part of a holonic system containing n holons (connected or not), the principal holon being the universe itself—understood not as a celestial body but as the maximal container.*

Stated this way the thesis is exhilarating and, I will argue, mostly false—but **instructively** false. The interesting question is never "is everything a holon?" (to which the answer is either *trivially yes* or *no*), but: **when does calling a system a holon say something that could be wrong?** That question has a precise, defensible answer, and locating it is the contribution of this paper.

### 1.2 Three readings, and the vacuity hazard named up front

The thesis admits three readings that must never be allowed to share credit.

- **Weak / lens reading.** *Any* system can be *described* as holonic under some part–whole decomposition. This is true and trivially so—under classical extensional mereology with unrestricted composition, every individual is a part of the universal fusion. It predicts nothing and excludes nothing.
- **Strong / ontological reading.** Every system *is* holonically organised, as an observer-independent fact, with the universe as the maximal holon. I will argue this is **coherent metaphysics, not science**, and is *refuted as a universal* by the counterexamples of section 4.
- **Epistemic / methodological reading.** "Systems" are black-box behavioural models we *carve* (map $\neq$ territory); "holonic" is a disciplined way of *modelling and composing* systems by their behavioural interfaces. This is the reading that survives—**only if** it pays the discipline of section 3.

The danger that haunts all three is **vacuity**: a universal predicate that applies to everything typically forbids nothing, and a framework that forbids nothing is a classification scheme, not a theory. This is not a hypothetical risk; it is the documented fate of the thesis's closest ancestors.

1. **General Systems Theory.** Universal in ambition, it never stated its falsifiability boundary and degenerated into a perspective. An influential early voice in the movement diagnosed the trap: "we always pay for generality by sacrificing content, and all we can say about practically everything is almost nothing" (Boulding 1956).
2. **Pancomputationalism.** Putnam's triviality result shows that *every* ordinary open system can be read as implementing *any* finite-state automaton, via a post-hoc partition of its states (Putnam 1988); "this system computes X" is thereby drained of content. "This system is a holon" shares the structure that drives that result—a post-hoc decomposition can always be found. (Whether the holonic case ultimately escapes where pancomputationalism cannot turns on a disanalogy developed in section 2 and section 3: near-decomposability is a *graded magnitude*, and Putnam's result bites hardest on *binary* mappings.)
3. **Integrated Information Theory.** A formal theory that assigns graded consciousness to all physical systems—Tononi's IIT (Tononi 2008)—carrying strong ontological implications, drew a published charge of "pseudoscience" precisely for unfalsifiable panpsychism (Fleming et al. 2023). "The universe is the maximal holon" is its structural twin.

A theory with a boundary is real; a theory with none is a tautology. The rest of this paper is the search for that boundary—the stance throughout being that the boundary is the point, not a threat to be explained away.

### 1.3 The claim, and the method

**The claim.** Stripped of the universal and ontological overreach, a single defensible thesis survives, and it is constructive:

> The smallest set of epistemic commitments under which a holonic claim has empirical content is three, and they are inseparable: a declared **partition** into modules, a declared **timescale** of observation, and a **contraction counterfactual** (perturbing one module leaves its siblings approximately invariant at short horizons). Pre-registration is their institutional implementation. This constitutive framework is **necessary** for a holonic (equivalently: near-decomposable) claim to be falsifiable rather than Putnam-trivial. Whether it is also **sufficient**—whether such commitments *completely* block triviality—inherits an open problem (the Putnam–Chalmers question, section 3); the paper states this and does not claim the converse. **Demarcation follows as a consequence**: a system for which *no* admissible partition yields the near-decomposability signature is not a holon, and gains no predictive advantage from holonic modelling.

Everything else in the paper either earns this necessary condition or tests it against the systems that break it.

**The method.** The framework is built first, from primary sources read first-hand; its *consequences*—what it forbids—are then tested against the hard cases that fix its boundary. Each refutation-boundary is reported as a constraint the framework must satisfy, not explained away; the surviving framework is stated as what it *forbids*; and where the evidence is deflationary, the verdict is deflationary.

**Roadmap.** Section 2 locates the one place a holonic claim acquires testable content—near-decomposability—and the partition wall built into it. Section 3 states the constitutive framework: the three commitments, their minimality, and the open sufficiency limit. Section 4 draws the consequence—what the framework forbids—and fixes the contraposition class. Section 5 and section 6 then supply the conceptual anatomy the framework presupposes: the containment-versus-communication layering, the three asymmetries of downward influence, the dynamical-versus-semiotic distinction (the only route by which a whole's downward influence survives Kim's exclusion, section 5.4), and the mereological home. Section 7 turns the framework on its own maximal object and shows the capstone ill-formed. Section 8 gives the verdict: a necessary condition, deflationary, with a hard core.

### 1.4 Contributions

This paper makes the following contributions.

- A **minimal constitutive framework** for the empirical content of holonic claims: three epistemic commitments—a declared partition, a declared timescale, and a contraction counterfactual—fixed before the data, with pre-registration as their institutional implementation. Claimed as *necessary only.*
- A **minimality** result (section 3.2): each commitment is required—no pair suffices—and no further commitment identified does additional logical work.
- A **synthesis** integrating Koestler's holon with Putnam-triviality via Simon's near-decomposability and Bousquet–Slotine contraction, locating the partition-presupposition as the wall it shares with the Free Energy Principle.

- A **conceptual anatomy** of holonic claims (section 5–section 6): the containment/communication layering, the three asymmetries of downward influence, the dynamical/semiotic distinction, and the reflexive capstone boundary.
- An **explicit dependency map** (section 8.2) of what the framework rests on versus what is hostage to open problems.

**What existed, and what this paper adds.**

| Ingredient | What existed | What this paper adds |
|---|---|---|
| Near-decomposability | Simon: a timescale-gap signature | a constitutive commitment for holonic content |
| Critical coupling $\varepsilon_c$ | Bousquet–Slotine: a threshold, partition given | a falsifiability gate, coupled to preregistration |
| Timescale-relativity | Weinberger: decomposition is grain-relative | bound into the (partition, timescale) unit |
| Anti-triviality | Chalmers: a counterfactual for implementation | relocated to partition admissibility |
| Arbitrary partition | FEP critics: the blanket is stipulative, post-hoc | the shared wall, closed by preregistration |
| Holon | Koestler: Janus, informal | the constitutive framework; dynamical/semiotic; the capstone diagnostic |

# 2 The structure↔dynamics hinge: near-decomposability

This is where the thesis stops being a re-description and acquires content that could be wrong. The bridge from "organised into parts and wholes" (a structural claim) to "behaves in a way a flat model cannot capture" (a dynamical, testable claim) is a single concept with a fifty-year pedigree: **near-decomposability**.

## 2.1 The one discriminating prediction — and a number

Simon (1962) gave the holonic intuition its one falsifiable signature. A *nearly decomposable* system is one whose interactions are far stronger within subsystems than between them, and from that structural fact Simon derived two dynamical propositions, verbatim: "(a) in a nearly decomposable system, the short-run behavior of each of the component subsystems is approximately independent of the short-run behavior of the other components; (b) in the long run, the behavior of any one of the components depends in only an aggregate way on the behavior of the other components."

This is the prediction a holonic description makes that a flat interaction graph does not: a **separation of timescales** — fast intra-module relaxation, slow inter-module coupling — observable as a spectral gap. It is genuine content: a system whose dynamics show no such gap is not nearly decomposable, and a holonic model of it buys nothing.

Two facts from the same source bound the thesis. First, **Simon declined the universal quantifier** the present thesis asserts: hierarchy is what complexity "frequently" takes, in "a large proportion" of cases — "at least half the truth," though "it may not be the whole truth." He even pre-diagnosed the immunisation trap as an *observation-selection* effect: non-hierarchic systems "may to a considerable extent escape our observation and our understanding." The universal "any" is not Simon's; it is the thesis's addition, and section 1.2's vacuity hazard is the price of that addition. Second, Simon's "hierarchy" is **not authority** — "the formal hierarchy exists only on paper" — which is the section 5.3 point arriving from the dynamics side.

A 2015 result sharpens the signature from a qualitative gap into a **critical threshold**. Bousquet and Slotine, combining contraction analysis with singular perturbation, prove that for inter-module coupling strength $\varepsilon$ there *exists* a **critical value** $\varepsilon_c$: when $\varepsilon < \varepsilon_c$ the coupled system contracts onto an $O(\varepsilon)$ neighbourhood of its *slow manifold* — the whole is, provably and within a bounded error, the stable sum of its parts — and when $\varepsilon$ exceeds $\varepsilon_c$ that guarantee fails, giving way to multiple equilibria and then instability as coupling grows. The demarcation becomes concrete: **a holon is a level at which intra-module coupling exceeds inter-module coupling and $\varepsilon < \varepsilon_c$.** The complementary repulsion regime ($\varepsilon > \varepsilon_c$), in which a system admits no contracting decomposition, is a *sharp threshold, in principle measurable* once the closed form of $\varepsilon_c$ is fixed — not a vague matter of judgement. (That closed form is reported here only at the level of its structure; it awaits confirmation against the source's primary typesetting and is **not relied on** — the argument uses the *existence* of the threshold, not its value.)

## 2.2 The wall built into the signature: the partition is presupposed

The signature is not free of the section 1.2 vacuity hazard — and the wall comes from the primary itself. Bousquet and Slotine take the **module partition as given, not derived**: their method *validates* a proposed decomposition; it does not *find* one. So "has a timescale gap" is not a property of a *system*; it is a property of a **(system, partition) pair**. That reproduces the post-hoc-partition freedom that drives Putnam's result: faced with a system, one may hunt for the grouping that makes the target signature appear. (The disanalogy that lets the holonic case escape where pancomputationalism cannot — that near-decomposability is a *graded magnitude*, not a binary mapping — is developed in section 3.)

The thesis's chief modern rival concedes the identical point. Friston's Free Energy Principle individuates things by Markov blankets, yet Friston himself calls a blanket "an implicit partition of states" that "…may sound a bit arbitrary," with "no privileged scale." The most rigorous universal frameworks and the holonic thesis share one wall: the decomposition is an *input*, and a universal predicate that quantifies over *all possible* inputs forbids nothing.

The way through that this paper takes — one option among more than one, but the one that keeps the thesis empirical — is to **re-quantify over partitions**: a system is holonic iff there *exists* a **system-admissible** partition at which the near-decomposability signature clears threshold, and *non*-holonic iff *no* admissible partition does. (This biconditional *defines* "holonic" relative to a fixed admissible class; it is distinct from the separate, open question of section 3.4 — whether the discipline *suffices* to block triviality.) (A realist alternative — that some partitions are causally privileged independently of the modeller — is a live but separately-defended position, left open here; see section B.) This makes the thesis explicitly **epistemic and partition-parametrised** — a claim about (system, admissible-partition) pairs, not about systems *simpliciter* — and it carries an asymmetry the method must own: *holonic* is confirmable by exhibiting **one** admissible partition that clears threshold, whereas *non-holonic* requires that **no** admissible partition does, a quantification over the entire admissible class. The negative verdicts on which the adversarial-demarcation method depend are therefore only as strong as the bound on "admissible" — which is why section 4.2 must supply that bound for the counterexample class. The criterion is non-vacuous only if "admissible" is fixed *before* the data, by a rule that is not "whatever makes the signature appear." That rule — pre-registration, given precise content by Chalmers's counterfactual constraint — is the subject of section 3, the constructive core of the paper.

# 3 The constitutive framework (the constructive core)

The diagnosis of section 2.2 was that "is a holon" reduces to "is near-decomposable under *some* partition," which — quantified over all partitions — forbids nothing. This section states the discipline that restores content, separates what that discipline establishes (a necessary condition) from what it cannot (sufficiency — an open problem the thesis *inherits* rather than solves), and positions the move against the prior art that shares its architecture.

## 3.1 The criterion

A holonic claim earns falsifiable content *only* when, and only to the extent that, it is made under three commitments fixed *before* the data:

1. **A pre-registered partition.** The decomposition into modules is declared in advance, drawn from a fixed admissible class, and may not be re-chosen post hoc to make the signature appear.
2. **A pre-registered timescale.** The temporal grain at which the near-decomposability ratio is read is declared *with* the partition — because near-decomposability is timescale-relative: the same variable is exogenous at one grain and endogenous at another (Weinberger 2020). Partition alone is insufficient — by the timescale-relativity result it underdetermines the signature; the **(partition, timescale) pair** is the unit of the claim.
3. **A contraction counterfactual.** The pre-registered partition must support the counterfactual that perturbing one module below the critical coupling leaves its siblings approximately invariant at short horizons (the Bousquet–Slotine $\varepsilon < \varepsilon_c$ regime, section 2.1). This is what makes the partition track a real interaction structure rather than a convenient relabelling.

Under these commitments the claim forbids an observation: a system that shows no timescale gap at the declared (partition, timescale), and whose modules do not contract-decouple, is *not* a holon at that grain, and a holonic model of it buys nothing. **The contraposition is the content** (section 4 fixes the class of systems that fall in it).

## 3.2 Minimality: each commitment is required, and no fourth does additional work

The three commitments are not a checklist; they are the minimal closure of the anti-circularity move. *Each is required*: dropping any one re-opens the freedom section 2.2 diagnosed, and each pair fails for a *distinct* reason.

- **Partition and timescale, without the contraction counterfactual,** is a *relabelling*: a grouping and grain can be declared that exhibit a gap by construction, without the gap tracking any real interaction structure. The counterfactual is what ties the signature to structure rather than to coincidence.
- **Partition and counterfactual, without a declared timescale,** is *underdetermined*: by the timescale-relativity result (section 3.1, Weinberger (2020)), the same variable is exogenous at one grain and endogenous at another, so the signature is not defined until the grain is fixed.
- **Timescale and counterfactual, without a pre-registered partition,** is *Putnam-gameable*: one hunts post hoc for the grouping under which the counterfactual happens to hold (section 2.2).

The three failure modes — non-tracking, underdetermination, circularity — are distinct, so no pair recovers what the third supplies. *And no fourth commitment, on the analysis here, does additional logical work.* The natural candidates reduce to the three: the choice of *observable* — which proxy measures the ratio — belongs to the *operationalisation* of the contraction counterfactual, not to the logical constitution of the claim, and is admissible only as a faithful operationalisation of it; a coupling-strength threshold is already the $\varepsilon < \varepsilon_c$ regime; a boundary or interface declaration is already part of the partition. This is a claim about the commitments identified here, not a proof that no fourth could exist: a reader who proposes one must exhibit a commitment that does logical work the three do not.

### 3.3 Why this is the right *kind* of fix: the Chalmers escape

The pre-registration move is not ad hoc; it is an instance of the standard escape from triviality results. Putnam's theorem bites because it permits a *post-hoc, material-conditional* mapping. Chalmers (1996) showed the escape: demand reliable, **counterfactual-supporting causal structure** — transitions that would hold counterfactually, localised in independent components. The contraction counterfactual of section 3.1 is exactly such a strong conditional, and the pre-registered partition is the localisation demand. The holonic thesis thus inherits a *recognised* anti-triviality strategy rather than inventing one.

One rebuttal bears directly on this case: Bishop (2002) shows counterfactual constraints can be satisfied trivially over a *finite window* — exactly where any real spectral-gap or contraction measurement lives. The graded-magnitude disanalogy (section 1.2, section 2.2) is what blunts it: Bishop's construction trivialises a *binary* FSA-trace implementation, whereas the near-decomposability ratio is a *continuous magnitude* that, measured over a finite horizon, can still come out at any value — so the finite-window trick does not collapse it to a forced yes. This does not *establish* sufficiency (section 3.4); it shows the contraction counterfactual is less exposed to the leading rebuttal than a binary implementation claim would be.

### 3.4 The limit: sufficiency is an open, contested problem

That counterfactual/causal structure is **necessary** for non-triviality is secure. That any such constraint is **sufficient** — that it *completely* blocks triviality — is, in the source debate, an open and contested question, and the balance of the verified literature tilts *against* sufficiency:

- Chalmers (1996) proposed the counterfactual-plus-localisation constraint as the block, yet in the same paper named two residual failure modes and called the account "imperfect as it stands", needing "some sort of revision or addition"; in his later reply (Chalmers 2012) he conceded a further "spatial independence" patch was required, one that "rules out the relevant trivializations but at the cost of ruling out some reasonable implementations."
- The direct rebuttal (Bishop (2002) — "counterfactuals cannot count") and the leading survey (Sprevak (2018)) hold that counterfactual constraints alone are insufficient; the mechanistic and semantic camps (Piccinini (2015); Rescorla (2013)) argue the same from inside the positive-account programme.
- That a *new* sufficiency criterion was still being proposed in 2024 (Anderson & Piccinini's physical-computational equivalence), and that the field's canonical survey treats the matter as an unresolved, escalating series (SEP, *Computation in Physical Systems*, 2024), are the decisive meta-signal: no constraint has been shown sufficient. A heterodox minority (Schweizer (2019)) argues the triviality argument is itself invalid — which *dissolves* rather than settles the question; hence *open-contested*, not cleanly closed.

The consequence for this paper is exact, and it is the one section 1.3 already draws: **the criterion is claimed as necessary, not sufficient.** The holonic thesis does not solve the Putnam-triviality problem; it **inherits** it. This is the precise sense in which the surviving core is *non-vacuous but hostage* — its full anti-triviality guarantee is held hostage to a thirty-year-open problem in the philosophy of computation. Sufficiency is therefore kept *as an open possibility, not an assertion*: the literature does not refute it, but neither does it earn it. Locating the thesis's open problem inside this active, named debate is itself a result — the holonic thesis's weakest joint is a *shared* frontier, not a private defect.

### 3.5 What is borrowed, and what is the advance

A claim of novelty must separate the *architecture* from the *operationalisation*.

**The architecture is prior art.** "A universal framework, disciplined by an applicability gate that something must pass to count, with a graded rather than binary boundary" is a move made independently in at least three traditions: Ladyman & Ross's *Principle of Naturalistic Closure* in ontic structural realism (a gate on what counts as a real pattern); Godfrey-Smith's *Darwinian populations* (disciplining a near-universal claim by placing systems in a graded parameter space instead of drawing a binary line); and — most sharply — independent voices inside the Free Energy Principle (critics and defenders alike), who arrive at the **identical partition-presupposition wall** this paper diagnoses: that assigning the Markov blanket is "ad hoc and follows the convenience of each case" (Di Paolo et al. 2022), rests on "delicate, non-fundamental stipulations" (Sánchez-Cañizares 2021), is "stipulative" by design (Ramstead et al. 2023), and needs extra premises to do any metaphysical work (the Pearl-vs-Friston distinction, Bruineberg et al. 2022). That four independent groups hit the same wall from inside the leading rival framework is strong corroboration that the wall is real — and it makes this paper's diagnosis a *synthesis*, not a discovery.

**The advance is the operationalisation.** What no surveyed source supplies is the specific move: (1) **pre-registration** of the partition *and* timescale as the anti-circularity device — every analogue in the FEP literature is *post-hoc* (deriving conditions to *check* a blanket: Heins and Da Costa (2022); detecting one algorithmically from data: Beck and Ramstead (2025); or conditioning on already-known dynamics: Hipólito et al. (2021)); and (2) the use of a **contraction threshold** — the *existence* of Bousquet & Slotine's $\varepsilon_c$ — as the falsifiability gate (a gate on existence, not a confirmed closed-form value; see section B), where Bousquet & Slotine themselves leave the partition *given* and never address the circularity that pre-registration closes. The defensible contribution is therefore narrow, and statable in one line: **a pre-registered, timescale-indexed, $\varepsilon_c$-gated demarcation for holonic claims — bridging a demarcation architecture with prior art in structural realism, graded Darwinism, and the FEP partition debate, and supplying the pre-registration and the quantitative gate those leave open.**

This locates the borrowing precisely. Simon (1962), Bousquet and Slotine (2015), and Friston (2019) independently reach the same slow/fast timescale-separation structure — none using the word "holon" — so the *falsifiable layer* is not this paper's invention, and four independent groups inside the Free Energy Principle hit the same partition-presupposition wall it diagnoses (above). What is this paper's is the operationalisation just stated, together with two contributions of clarity it makes possible: a **causal taxonomy** (authority / ascending / constraint; dynamical versus semiotic, section 5.4) and a **containment logic** (the maximal fusion $U$ under contested universalism), both developed in section 5–section 6. What results is a criterion for the *empirical content* of holonic claims — the conditions under which such a claim has any — with demarcation following as a consequence, not a body of new predictions.

# 4 The objections, and the contraposition class

The criterion of section 3 is only as good as the objections it survives and the systems it excludes. This section runs the four hardest objections and fixes the **contraposition class** — the systems the criterion calls *non*-holonic — because, on the adversarial-demarcation method, that class is the thesis's empirical commitment.

## 4.1 The vacuity / Popper–GST wall

The vacuity hazard named in section 1.2 — that "every system is a holon" forbids nothing, the documented fate of General Systems Theory and the structural twin of Putnam's triviality — is the wall the framework must survive. It survives *only* by the pre-registration discipline of section 3, whose escape route is a recognised one (the Chalmers counterfactual constraint, section 3.3), with the caveat established there that its *sufficiency* is an open problem the thesis inherits.

The model of content-earned-by-exclusion is **autopoiesis**: Maturana and Varela offered self-production as a *necessary and sufficient* condition for life — content won precisely by what it excludes (a factory is *allopoietic*, not alive; crystals and flames are the standard textbook cases). The holonic criterion aspires to the same shape: a predicate that says "no" to clear cases. (A caveat from the source: autopoiesis's *sufficiency* half is itself contested — Bénard cells and the authors' own later reformulations strain it. The model to imitate is the *exclusion discipline*, not the claim that the exclusion is perfect — the section 3.4 lesson restated.)

## 4.2 The contraposition class: systems that resist decomposition

The criterion earns content by naming systems it calls non-holonic. Three are confirmed:

- **Quantum non-separability.** Entangled states do not factor into subsystem states; the "parts" are not semi-autonomous. The near-decomposable reading is *false* here. (One nuance keeps it instructive: entanglement is the *extreme of the integrative pole* — whole-prior-to-parts — so it refutes the autonomous-parts reading while instantiating the holistic one. Caveats from the source: non-factorisation is interpretation-relative, and Healey's pragmatist account reproduces the correlations with no metaphysical holism at all. The refutation of "ANY system" stands regardless.)
- **Fully-developed turbulence.** The inertial-range energy flux is scale-*local* (Eyink–Aluie) — so the naïve "non-local cascade" story is wrong — yet small scales are *not statistically autonomous* from the integral scale: intermittency and persistent anisotropy keep them sensitive to large-scale conditions. A clean Simon decomposition into near-independent scales fails. Ubiquitous, classical, and exactly where the thesis's own contraction formalism predicts decomposition breaks.
- **Self-organised criticality and scale-free flocks.** Scale-free correlations, *no characteristic scale* (Cavagna et al. on starling flocks) — a paradigm "organised whole" with no timescale gap to find. (Attribution: Cavagna establishes the scale-free/critical fact; "no timescale gap" is the inference from it.)

The surrendered territory is large — plausibly the quantum foundation, the generic regime of continuous classical matter, and much of criticality phenomenology. That is the price of "any." (One *attempted* counterexample fails: high-dimensional chaos is a *tractability* barrier — Takens embedding still reconstructs the attractor — not ontological resistance.)

A single system illustrates — and a worked measurement *sharpens by correcting* — how partition-relative the verdict is (section 2.2). Take the complex Ginzburg–Landau equation, the canonical amplitude-equation for such media (the Belousov–Zhabotinsky reaction is the paradigm chemical instance), in two regimes: Benjamin–Feir-*stable* (coherent plane waves) and *defect-turbulent*. Under a pre-registered spatial-block partition with a correlation-length signature (section D), the measured verdict runs *opposite* to the naive "organised = decomposable" intuition: the coherent regime is long-range-ordered (correlation length $\xi \approx 0.8$ of the block size) and is therefore **not** block-decomposable, whereas defect turbulence has a short $\xi \approx 0.12$ of the block and so **is** near-decomposable into quasi-autonomous blocks. The lesson is *not* that turbulence is "holonic" and order is not; it is that holon-hood is a property of the *(system, regime, partition, signature)* tuple — so sharply that it can invert the intuitive reading. (This also locates the three members above precisely: fluid turbulence and SOC sit in the contraposition class for a *different* reason — cross-*scale* coupling and scale-freeness — not the spatial-block decomposability measured here.)

### 4.3 What the bounded thesis forbids (the pre-registrable predictions)

A flat description predicts no privileged timescale gap and no contraction-bounded module independence; the holonic thesis predicts both, and four more — each pre-registrable and risky:

1. **Timescale-gap (master falsifier).** A holon must show a measurable spectral gap (intra-module relaxation faster than inter, by the critical ratio) at the declared (partition, timescale). *Risk:* a robustly modular, self-maintaining system with a flat spectrum, insisted to be a holon → falsified.

2. **Contraction-perturbation (the anti-triviality counterfactual).** Perturb one sub-holon below $\varepsilon_c \rightarrow$ siblings stay approximately invariant at short horizons. *Risk:* short-horizon propagation reorganises siblings despite apparent modularity.

3. **Regime-shift.** Past a disturbance threshold, lower holons increase autonomy and re-topologise control, and the heterarchic transient performs *worse* than the stationary hierarchy under normal load. *Risk:* the flat/heterarchic regime performs equally or better.

4. **Channel-capacity bound.** Ascending feedback that modifies the parent is bounded by the interface's measured information capacity (Ashby's Law of Requisite Variety). *Risk:* a sub-holon reorganising its parent through a channel whose capacity is below the reorganisation's information content.

5. **Non-reducibility-of-constraint (riskiest, aimed at Kim).** The parent's constraint is a *non-integrable* (semiotic) constraint whose functional description is lost under micro-redescription. *Risk — and the likely failure mode for dynamical holons:* if every constraint is ordinary integrable physics, downward constraint-causation collapses to redescription. This is why section 5.4 relocates the live fight to semiotic holons.

The contraposition test is the headline: *a system lacking the contraction-plus-timescale signature under any admissible (pre-registered) partition is not a holon and gains no predictive advantage from holonic modelling.* That is what the thesis forbids — and what makes it more than a perspective.

# 5 What a holonic thesis must commit to: the causation topology

Before the thesis can be tested it must be made precise about a question it usually leaves implicit: **in which directions does influence run between a holon and the levels above and below it?** Holism is heard, by turns, as a pyramid — higher levels command lower ones, which merely execute — and as its opposite, a flat web in which everything affects everything. Neither is the holonic claim. Fixing what *is* the claim forces a distinction the literature routinely blurs, and that distinction turns out to defuse the thesis's hardest-looking commitment.

## 5.1 Four topologies

Four positions are on the table, and they are not interchangeable.

- **The pure pyramid.** Strict top-down determination: lower levels are fixed by higher ones and never determine them. This is the reading the present thesis explicitly rejects — and so, decisively, did Koestler.
- **Koestler's Janus-faced holon.** Every holon (Koestler 1969) "display[s] both the autonomous properties of wholes and the dependent properties of parts" — the *Janus phenomenon* — its self-assertive (whole) and integrative (part) tendencies held in dynamic equilibrium. Koestler rejects *both* atomism and holism: "parts and wholes in an absolute sense do not exist in the domains of life." And he severs hierarchy from authority: "the canon determines the rules of the game, strategy decides the course of the game" — the parent fixes the *possibility-space* (the boundary conditions), the child is strategically autonomous within it. This canon-versus-strategy split anticipates the authority/causation distinction below.
- **Heterarchy.** McCulloch (1945) showed that the value-orderings realised in a nervous net can be cyclic — A preferred to B, B to C, yet C to A — and argued that this is not an inconsistency but "consistency of an order too high to permit construction of a scale of values." Peer influence can form cycles that a tree cannot represent.
- **Tangled hierarchy.** Hofstadter's (1979) level-crossing self-reference, in which no element holds a unique level. (Not load-bearing below; see section B for its status.)

A Koestlerian holarchy is, structurally, a **tree**: one parent per holon. A tree cannot represent cyclic peer influence (heterarchy) or level-crossing loops (tangled hierarchy). The thesis, however, appears to want both at once — a single maximal tree topped by "the universe," *and* a network of laterally communicating holons. Read as one relation, that is a contradiction.

## 5.2 The two-layer resolution

It is not a contradiction once we see that the two intuitions are **two distinct relations over the same holons**:

1. an acyclic **containment** order — parthood: antisymmetric, a tree in the one-parent case, with the maximal whole on top; and
2. a cyclic **communication / causal** network — heterarchical, lateral, apex-less — through which holons influence one another.

Parthood's antisymmetry *forbids* it from doubling as the lateral-influence relation; conversely the communication graph carries cycles that a parthood tree cannot. Conflating them manufactures the apparent paradox; separating them dissolves it. The two-layer picture is not a convenience: it is already instantiated in a built holonic system (section E), where an acyclic containment hierarchy and a lateral stigmergic field are *separate* structures. Which layer dominates is, moreover, *dynamical*: if a system's sub-holons only differentiate downward, the containment stays a tree (and tops out in the clean maximal fusion $U$ of section 6); if their couplings also *merge* — siblings re-joining, a holon acquiring more than one parent — the containment ceases to be a tree, collapses toward the lateral network, and the clean $U$ is lost with it. The boundary between the two layers is therefore something a system's dynamics can cross. The lateral layer matters immediately below — it is where the thesis's most cherished example actually lives.

### 5.3 Three asymmetries the literature conflates

The decisive clarification is that "downward influence" names **three** different relations, only one of which is philosophically contested.

1. **Authority / boundary-condition asymmetry.** The parent fixes the possibility-space — Koestler's *canon* — within which the child acts. This is uncontested: a constitutive fact about constraint, orthogonal to causal efficacy. (It will reappear in section 5.4 as the candidate home of the *only* surviving downward route.)

2. **Ascending causation.** Parts affect the whole. Uncontested; it survives every wall trivially.

3. **Descending causation.** The whole exerts influence on its parts *as a distinct cause*. This is the only contested direction — the target of Kim's exclusion argument (section 5.4).

The payoff is clarifying for the thesis rather than damaging. Holism's defenders tend to present "the whole acts back on the parts" as one bold, embattled claim; once the three relations are separated, most of what they mean — authority and ascending feedback — is uncontested, and only *strong descending causation* is genuinely at risk. In particular, the thesis's flagship illustration of bidirectionality — a subordinate node modifying the very node that governs it, through documented feedback — turns on *which layer the channel occupies*. Insofar as the link is **lateral** (the communication network), it is uncontested horizontal influence across a real, capacity-bounded channel — the safest case. Insofar as it runs **parent-ward** (the containment tree), it is *ascending* causation onto a whole — still uncontested *as ascending*, though it meets the exclusion question from the harmless side. Either way it is not the hard case (strong *descending* causation), and section 5.4 shows the hard case survives only in a special, *semiotic* form. Naming which asymmetry — and which layer — one is asserting is therefore not pedantry: it is what keeps the thesis from claiming, and having to defend, more than it needs.

### 5.4 Kim's causal-exclusion wall

The hardest wall for any *bidirectional* reading is Kim's exclusion argument. Four principles cannot all hold: a higher-level property is causally efficacious; the physical domain is causally closed; the higher property is irreducible; and no effect has two sufficient causes (no systematic overdetermination). Edwards's dictum sharpens it — "sufficient synchronic determination relations exclude causal relations" — so the synchronic constraint of a whole on its parts is exactly the fenced-off zone, not a safe harbour. And the **drainage** generalisation bites: if a

macro-property is excluded by its sufficient micro-base, the same reasoning drains causation down every level to microphysics.

The surgery section 5.3 prepared now pays off. *What dies:* descending constraint as an ontologically *distinct* causal channel. *What survives:* ascending causation (trivially); horizontal/lateral influence (the flagship example); authority / boundary-condition setting (Kim is about efficacy, not who fixes the possibility-space); and *epistemic* bidirectionality (the macro order-parameter carries real predictive information at the control-relevant grain). The drainage argument is also the thesis's best *defensive* card: taken to its conclusion it would erase chemistry's and biology's causation too, so rejecting the *universality* of the exclusion principle is more reasonable than accepting the drainage — a defence by *reductio*, not by winning the production claim.

This forces a taxonomy that is content, not retreat: **dynamical holons** (contraction/timescale-based) are Kim-reducible and ascending-only; the *only* live route for irreducible downward influence is the **semiotic holon** — a rate-independent symbolic constraint that selects among configurations the dynamics underdetermine (Pattee's non-integrable constraint, his epistemic cut). Pattee is explicit this is *epistemic, not ontological dualism*, and full escape from Kim remains open: it turns on whether symbolic selection *adds* causal specification or is micro-determined (the degeneracy gap, an open question). The distinctive bidirectional claim must therefore be confined to *semiotic* holons — a sharp, falsifiable restriction.

An audit of the built system (section E) finds this surviving route *empirically instantiated* — and the finding tightens the restriction rather than loosening it. A rule there canonicalises distinct byte-encodings of the same text (UTF-16 LE/BE, UTF-8 with and without byte-order mark) to one form: a genuine *rate-independent, conventional* symbol-handling operation, collapsing representation-equivalent inputs by a rule the dynamics do not fix.[1] But it is **generic**: encoding canonicalisation is ubiquitous to *any* software that interprets externally-encoded data, it is not emergent from the holonic architecture, and it is dynamically inert — it selects a representation, it does not give a whole an irreducible downward efficacy over its parts. So the only Kim-surviving route exists in the system, yet does not *distinguish* what is holonic from what is ordinary data interpretation. The restriction of section 5.4 is confirmed, not relieved: the live route is real *and* generic.

# 6 The formal home: mereology, not set theory — but weaker than it looks

The thesis carries a formal commitment: that the right container for "the universe as the maximal whole" is **mereology** (the theory of parthood), not set theory. The instinct is correct on its negative half and weaker than first stated on its positive half — a distinction worth getting right, because the capstone stands or falls on whether the maximal whole is a legitimate object at all.

## 6.1 Parthood versus membership, and the corrected status of U

Mereology and set theory differ in exactly the way the thesis needs. Parthood (Varzi 2019) is reflexive, transitive, and antisymmetric, and — unlike set theory — "mereology is not committed to the existence of abstracta"; membership ($\in$), by contrast, is not transitive. So the **negative claim holds**: set theory is the wrong home for a maximal container, because its universal class

[1] Observed by invoking the existing `decode_to_utf8` on four distinct byte representations of one string; all map to the identical output, a non-byte-identical control maps elsewhere. Audit probe and commands in the repository (`pslab_run/`); a claim until re-run.

$V$ is provably *not a set* (Russell's paradox) — not an object of the theory at all — whereas a mereological "universe" is at least a candidate individual.

The **positive claim, however, is weaker than the thesis assumes**, and this is a correction the verification forced. It is tempting to say the maximal fusion $U$ is a *theorem* of mereology. Varzi is explicit that this is too strong: $U$ exists *only* if one assumes a universal entity or adopts **unrestricted composition** — and, in Varzi's words, "Both assumptions, however, are controversial." Mereological nihilists deny exactly this. So "the universe as a maximal individual" is a **contested axiomatic choice**, not a forced result. The capstone cannot lean on $U$ as if mereology handed it over for free; it must own the universalist commitment that buys it.

### 6.2 Connection is not parthood; extensionality is contested

Bare classical mereology is, in any case, too thin to host a *holon*, for two reasons.

First, the holon's defining feature is a **boundary / interface**, and parthood cannot express it. Two adjacent rooms share no part yet are *connected*; bare parthood has no way to say so. Expressing the holon's individuating relation requires adding a topological primitive — **connection** — on top of parthood (mereotopology, in the Whitehead lineage Varzi surveys). The load-bearing primitive for holons is therefore mereotopological connection, not bare parthood.

Second, a tempting argument that mereology is *inadequate* — the failure of **extensionality** (the statue and the clay share all their parts yet seem to differ) — turns out to be **contested**, not decisive. Varzi shows classical mereology has standard replies (such cases reduce to diachronic identity or to *de dicto / de re* modal scope, and so are "not genuine counterexamples to extensionality in classical mereology"). Extensionality is, then, *contested*: the holon's structure-dependent identity sits with those who find extensionality inadequate, but the classical mereologist has a defence. The thesis may not bank extensionality's failure as established.

### 6.3 The transitivity-severance pincer

The hardest mereological wall is one Varzi states in his own words: functional or organisational "part" is **not transitive**. "A handle can be part of a door and the door of a house, though a handle is never part of a house; my fingers are part of me and I am part of the team, yet my fingers are not part of the team." Varzi resolves the puzzle by ruling that the functional relation (call it $\varphi$-part) is "not 'part'" in the strict mereological sense — precisely so that mereological parthood can stay transitive.

But the holon's level-individuating relation *is* the functional one. That forces a pincer with no clean branch:

- **(a)** Identify holon-part with mereological part. Then it is transitive and **level-blind**: the holarchy collapses into the flat part-order that the thesis was supposed to out-predict.
- **(b)** Identify holon-part with the functional relation. Then it is non-transitive and **non-mereological**, and it does *not* fuse into $U$; "mereology is the holon's home" becomes a category error.

Classical mereology can host the holarchy's de-organised *material shadow*, but not the holarchy *as a holarchy*. A tempting third branch — a level-indexed or restricted-transitivity parthood, mereological yet not level-blind — does not escape: level-indexing reintroduces exactly the organisation-relative relation that branch (b) isolates, so it is branch (b) in mereological clothing. The *stratified stack* endorsed in section 6.4 is accordingly the explicit form of branch

(b) — a layered structure whose level-individuating relation is, and is admitted to be, non-mereological — not a refutation of the pincer. (An independent strain confirms the cost: an absolutely general $U$ clashes with ZFC, because unrestricted fusion yields strictly more objects than there are sets — a cardinality result reported by Yao (2023), on an argument due to Uzquiano. Two escapes exist — restrict the quantifiers to urelements, the non-set individuals, or revise the fusion axioms — but either way the absolutely-general $U$ is not free.)

### 6.4 Verdict on the formal home

The negative claim **holds**: set theory is the wrong container. The positive claim **demotes** to a *stratified stack* — containment parthood (the section 5.1 tree, $U$-topped only under contested universalism) plus mereotopological connection (the boundary/interface) plus organisation-dependent identity plus a *separate* causal/communication relation (the section 5.1 network, which parthood's antisymmetry forbids from *being* parthood).

The section 6.3 handle shows why that separate relation is indispensable. The handle parthood severs from the house — never a *part* of it — can still *connect* to it through a different relation entirely: as a **sign**. A familiar handle, met again elsewhere, can stand for the whole house (a sign relation in Peirce's sense — sign, object, interpretant; an *associated* element evoking the whole by **metonymy**, fitting precisely because the handle is *not* a part of the house but contiguous with it — this is association, not the part-for-whole of synecdoche; a cue completing a remembered pattern). That relation is real but *semiotic*, not mereological: it lives in an interpreter's model — a stranger reads no house in the same handle — it is *lossy* (the handle evokes the house, it does not contain or specify it), and it is not a causal coupling between the objects (perturbing the handle leaves the house untouched). It is precisely the rate-independent symbol of section 5.4, and it is exactly what parthood's antisymmetry cannot be.

The two slogans the thesis opened with — "holon-part = mereological part" and "the universe is the maximal *holon*" — do not survive intact: $U$ is the maximal *concrete fusion* — antisymmetric, connection-blind, explanatorily empty — the **least** holonic object in the framework. That is not a side-remark; it is the seed of the capstone's collapse.

## 7 The capstone, and the strong reading

The thesis's most ambitious claim — that the universe is the *principal holon* — is also the one place the framework turns on itself. This section argues the capstone is a category error on the holon's own definition, isolates the only "universe" claim that survives, and reclassifies (rather than dismisses) the strong ontological reading.

### 7.1 Why "the universe is the principal holon" is a category error

A holon, on Koestler's defining account (section 5.1), is Janus-faced: a whole *and* a part, displaying the autonomous properties of wholes and the dependent properties of parts at once. The totality has, by definition, no parent and no exterior — it is whole-only. It therefore possesses just one of the holon's two faces and is, on the framework's own terms, *not a holon*: it is the root whole, the maximal mereological fusion $U$ (section 6.4), which the same analysis showed to be antisymmetric, connection-blind, and explanatorily empty — the *least* holonic object in the framework, not the principal one.

The point is structural, not merely definitional, and it indicts the empirical layer too. Every component of the surviving framework is constitutively boundary-based: Simon's near-

decomposability is a *ratio* of intra- to inter-block coupling, Ashby's black box is defined by an input–output boundary, Friston's blanket is an inside/outside partition. The totality has no exterior — there is, in the cosmologist's phrase, only one universe, with "nothing else similar to compare it with" (Smeenk and Ellis 2017) — so it has no interface, no blanket states, and an *undefined* near-decomposability ratio (no inter-block term to form). The physics makes this concrete: the Wheeler–DeWitt equation, $\hat{H}|\psi\rangle = 0$, freezes the universal wavefunction precisely because there is no external clock relative to which the whole could evolve. Applying the thesis's own falsification diagnostic to the totality returns *undefined*, not *true* — a self-administered *modus tollens* on the capstone. The verdict is exactly as strong, and as bounded, as its premise: it refutes the capstone *as a holonic claim* — one that takes Koestler's Janus definition as constitutive — and is silent on non-holonic monisms that deny that definition, which section 7.3 grants are coherent (section 8.2).

### 7.2 The only surviving "universe" claim

What survives is narrow and must be stated narrowly: "the universe is the maximal holarchy" holds *only* as a static mereological claim about the *parts* — that all holons and their couplings nest within one maximal fusion — and *never* as a dynamical or inferential predicate of the whole. The discipline this paper adopts and recommends is a rule: state every "universe" claim as the static maximal-fusion reframe; never attribute interface dynamics, inference, or a near-decomposability signature to the totality itself. The capstone is not softened into vagueness; it is replaced by the one claim that is true.

### 7.3 Reclassification, not dismissal

It would be too quick to call the strong ontological reading meaningless. "The whole is prior to its parts" has respectable, peer-reviewed homes: Schaffer's priority monism (the cosmos as the one fundamental object, with metaphysical explanation running downward; (Schaffer 2010)), ontic structural realism (relations prior to relata; Ladyman and Ross (2007)), and the processual revival in philosophy of biology. Each, however, describes *itself* as metaphysics rather than empirical science — none forbids an observation or yields a prediction. The correct verdict is therefore *reclassification*: the strong reading is coherent, well-housed metaphysics — not nonsense and not science. "Vacuous as science" survives; "meaningless" does not. The deflation removes the ontological intuition from the scientific ledger; it does not license contempt for it.

### 7.4 Why no rescue restores the part-face

Several natural strategies try to give the totality back its missing part-face. Each fails in an instructive, convergent way — and the convergence is itself the result.

- **Self-reference.** The universe contains a model of itself (a strange loop; the self-observing cosmos). One might press the strongest form: this self-connection is no *trivial* identity but a *transforming* recursion, and its stable self is an **eigenform** — the fixed point of the operator of self-observation (Kauffman's $O(A) = A$; von Foerster's "objects as tokens for eigen-behaviors"; (Foerster 1981; Kauffman 2003)) — "the same" only as the fixed point of a recursion that differs at every step; iterated, it becomes an unbounded layering of partial self-interpretations. Granted in full — and it still does not reach the totality: a self-model is a *proper part* (the map, not the territory), and the eigenform and *every stage and layer* of its recursion are themselves *interior* parts; the loop connects parts to parts. It restores the part-face only in the *representational* register — the surviving epistemic reading plus

a metaphysics of self-reference, not a dynamical holon. Mereologically, self-parthood is only the trivial reflexive case ($x \leq x$), not Koestler's "proper part of a *larger* whole."

- **Cosmological expansion.** Expansion changes the *interior's* degree of decomposability over cosmic time (an early, hot, strongly-coupled state → structure formation → late-time fragmentation) — a genuine enrichment of the bounded thesis, since near-decomposability is then *epoch*-relative, extending its timescale-relativity (section 3.1). But the metric expands; it does not expand *into* anything, so it creates no exterior — and accelerating expansion in fact *removes* reachable regions beyond the cosmological horizon. The no-exterior result is untouched.
- **Succession or multiplicity.** Speculative programmes (eternal inflation; conformal cyclic cosmology) in which one universe begets others either embed it in a larger multiverse — merely relocating the maximal whole up one level — or generate causally disconnected successors, which are *more* whole-only totalities, not parts of one another. A set of behavioural duplicates is a *class* (members sharing a property), and a class is neither a whole nor a holarchy: its members are not parts of each other.

Every strategy lands in the same place from opposite directions: each enriches the bounded thesis *for proper parts* — often with a genuine connection to frontier physics or metaphysics — and none makes the totality a holon. The totality is the unique object whose boundary-relation is degenerate, having no outside to relate to, which is exactly *why* a boundary-based framework cannot grip it. That is not a failure of the thesis; it is the thesis correctly reporting its domain of application: **proper parts with a genuine exterior, never the whole.**

# 8 Verdict: deflationary, with a hard core

The verdict is deflationary: most of the universal thesis does not survive, and what does is smaller, harder, and partly borrowed. The residue is real, and locating it precisely is the contribution.

**What survives.** A single, bounded, epistemic thesis: *holonic is a disciplined way to model and compose systems by their behavioural interfaces, and a holonic claim has falsifiable content only when its partition and timescale are pre-registered and its partition supports a contraction counterfactual.* It applies to **proper parts with a genuine exterior**; it is non-vacuous (it forbids models — the contraposition class of section 4.2); and it is necessary-but-not-(yet)-sufficient, its full anti-triviality guarantee hostage to an open problem in the philosophy of computation (section 3.4), claimed as such rather than papered over.

**What is dead or empty.** Three things. (1) The **capstone** — "the universe is the principal holon" — is a category error on the holon's own definition (section 7): the totality is whole-only and boundary-less, the least holonic object, not the principal one. (2) The **"complex dynamical system" disjunct** adds nothing: "dynamical system" is trivially universal (a constant qualifies) and "complex" has no canonical measure (the proliferation of dozens of mutually non-equivalent measures), so the disjunct is vacuous, false, or unfalsifiable depending on the reading — its only non-empty content is Simon's near-decomposability under another name. (3) The **strong, universal-ontological reading as science** — refuted as a universal by section 4.2, and deriving nothing it could be tested by.

**What is reclassified, not dismissed.** The strong reading *as metaphysics* is coherent and well-housed (section 7.3): priority monism, structural realism, processual biology. It forbids no observation — which bars it from the scientific ledger, not from sense.

**The contribution, without inflation.** I have not produced a new prediction: the falsifiable layer is Simon's, Bousquet–Slotine's, and Friston's, none of whom needed the word "holon"

(section 3.5). I have done two things instead. First, a **bridge** — forcing the holonic/Koestlerian tradition, popular and broad and often informal, into contact with the rigorous demarcation apparatus (near-decomposability, contraction, causal exclusion, the Free Energy Principle's partition debate) and reading off what the contact implies. Second, a **constructive criterion** — the pre-registered, timescale-indexed, $\varepsilon_c$-gated demarcation of section 3, whose specific move (pre-registration as the anti-circularity device, plus a contraction-threshold gate) no surveyed source supplies, every Free Energy Principle analogue being post-hoc. The price of admission is the word **"any."** The surviving thesis is a method, not a metaphysics; and a method that names its own exclusions is the opposite of the General Systems Theory it could have become.

**What the holonic framing adds over bare near-decomposability.** A fair challenge follows: if the falsifiable content is pure near-decomposability, has the paper merely re-labelled near-decomposable systems as holons — the very re-description it convicts GST of? The answer concedes the *empirical* layer is near-decomposability's, and locates the holonic framing's distinct work in three conceptual contributions a flat account does not make: the **two-layer** separation of an acyclic containment order from a cyclic communication network (section 5.2); the **dynamical-vs-semiotic** taxonomy that isolates the one place downward causation survives Kim (section 5.4); and the **Janus diagnostic** that turns the framework on its own maximal object and shows the capstone ill-formed (section 7). These are contributions to *structure and clarity*, not to prediction — unavailable to a near-decomposability account that lacks the whole-and-part vocabulary. Where the framing adds no predictive content the paper says so; where it adds structural content it is named.

**What this paper does not claim.**

- *Not sufficiency.* It does not solve, or claim to solve, the Putnam–Chalmers triviality problem (section 3.4); the framework is necessary only.
- *Not a constructive admissibility rule.* It does not supply a rule that fixes the admissible partitions non-circularly; it requires only that such a rule be fixed before the data, and marks the open problem (section 2.2).
- *Not that the strong reading is meaningless.* The universal-ontological reading is coherent metaphysics (section 7.3) — excluded from the scientific ledger, not from sense.
- *Not empirical corroboration.* The two worked demonstrations (section D; the self-falsification of section E) test the *method*, not the thesis; neither is independent evidence that any system is holonically organised.
- *Not a new result in dynamics.* The contraction threshold $\varepsilon_c$ is Bousquet–Slotine's; the falsifiable layer is Simon's. The contribution is constitutive and conceptual, not predictive.

### 8.1 Three registers, one with teeth

The framework can be formalised in three registers that must not be conflated. The *predictive* register — Simon's spectral gap and Bousquet–Slotine's $\varepsilon_c$ for a pre-registered (partition, timescale) — has teeth but is *local*, never global. The *universal* register, one mathematics for "everything," forbids nothing and collapses into vacuity; the Free Energy Principle (by its own critics) and the self-falsification of section E are its two existence-proofs. The *structural* register — a category of holons and behavioural interfaces over a mereological order of containment (section 6) — faithfully represents the whole vision but asserts no universal claim and derives no prediction: logically true, empirically empty. No register is *both* global and predictive, and the framework's content is precisely the discipline that says so.

## 8.2 Where the argument is weakest

**What rests on what.** The contribution separates into a secure part and a part hostage to open problems:

| Secure (independent of open problems) | Hostage to an open problem |
|---|---|
| The framework as a *necessary* condition: it disqualifies post-hoc, undisciplined holonic claims | That the commitments *suffice* to block triviality (Putnam–Chalmers, section 3.4) |
| The minimality of the triple (section 3.2); the conceptual anatomy | The system-level *non*-holonic verdict, which needs "admissible" bounded (section 2.2) |
| The synthesis (the shared partition wall) | The quantitative gate's exact value (the closed-form $\varepsilon_c$, section B) |

The load-bearing soft spots, named rather than hidden:

1. **The criterion is hostage (the deepest one).** Its sufficiency rests on the unsolved question of whether a counterfactual/causal constraint completely blocks Putnam-triviality (section 3.4) — and, relatedly, on whether "admissible partition" can be fixed in non-question-begging terms without smuggling in the semantic notions the structuralist account was meant to avoid (a worry Rescorla (2013) presses for computation generally). Should that thirty-year-open problem resolve *against* the constraint, the open sufficiency question is closed negatively; the criterion's status as "necessary only" — the status it is claimed under here — is unaffected, and it remains non-vacuous as such.

2. **The negative verdicts are harder than the positive ones.** "Holonic" is confirmed by exhibiting *one* admissible partition; "non-holonic" requires that *no* admissible partition works — a quantification over the whole admissible class (section 2.2). The contraposition class on which the method's content depends is therefore only as secure as the bound on "admissible." More sharply: because fixing "admissible" non-question-beggingly is the unsolved problem of section 3.4, the criterion's non-vacuity is demonstrated *in practice* only partly. Appendix D now exhibits one fully-worked, pre-registered, non-circular near-decomposability *verdict* on a concrete wave medium — but it is partition-*relative* (a positive, block-decomposable verdict in one regime, not-decomposable in another), not a confirmed *exclusion* under every admissible partition; the section E null, separately, tests a nonlinearity sub-claim, not holon-hood. A worked *confirmed exclusion* from holon-hood (no admissible partition succeeds) remains the first item of further work, and section 4.2's exclusions are arguments, not exhaustive proofs.

3. **The empirical anchor is thicker, but still bespoke.** Section E is now a four-run program — multi-regime overload calibration, a positive control, a real-latency injection, and an identifiability diagnosis — rather than a single null. The added stringency buys a sharper negative (the holonic term is *unidentifiable*, not merely zero) and a measured instance of the open underdetermination problem (section 3.4). But it is still *one bespoke system* testing a *nonlinearity sub-claim*, not independent corroboration of the thesis and not a confirmed exclusion from holon-hood. The anchor is heavier; it is not broader.

4. **Verification is real but not exhaustive.** Several load-bearing sources were confirmed at the abstract/landing-page level (V2), not read in full (Appendix B); the headline theses are confirmed, the supporting arguments are not all first-hand.

5. **The capstone verdict is definition-relative.** It follows from taking Koestler's Janus definition as constitutive (section 7.1); a priority monist who rejects that definition is not thereby refuted, only relocated to the metaphysics section 7.3 grants is coherent.

None of these is fatal; each is a place a careful reader should push, and several are live research directions rather than defects. Naming them is the same discipline as the rest of the paper: a boundary stated is a boundary owned.

# A Provenance (context of discovery)

The arguments of this paper stand entirely on the technical apparatus of the body; none of them depends on the route by which the thesis was reached. For transparency only: the thesis originated in the author's systems-design practice and was developed by mapping each informal intuition to its precise counterpart in the literature, then keeping only what survived first-hand verification. That working crosswalk, and the audited claim ledger behind every citation, are maintained as separate companion documents — the *context of discovery*, not the *context of justification* — and are not required to assess the argument.

# B Verification ledger

Every external source below was checked first-hand within this project. Its **verification level** follows the project convention: **V1** = the primary was read and the cited fact confirmed first-hand; **V2** = the primary was reached but the cited detail comes from its abstract/landing page; **V3** = secondary only, used as a lead and never load-bearing. The full audited evidence base — one row per claim, with the section it serves — is the companion `HOLONIC_THESIS_CLAIM_LEDGER.md`; this appendix lists the paper's load-bearing citations and their levels so a reader can confirm that none is fabricated.

**Read first-hand, fact confirmed (V1).** Simon (1962) (*Architecture of Complexity*); Chalmers (1996) (*Does a Rock Implement Every FSA?*); Anderson and Piccinini (2024) (*Physical Signature of Computation*); SEP *Computation in Physical Systems* (Piccinini 2024); Schweizer (2019); Di Paolo et al. (2022); Sánchez-Cañizares (2021); Ramstead et al. (2023); Hipólito et al. (2021); Schaffer (2010); Eyink and Aluie (2009); Cavagna et al. (2010); Kauffman (2003) (eigenform — the object as fixed point of a recursion, $O(A) = A$); Peirce's triadic sign relation (SEP *Peirce's Theory of Signs*, quoting Peirce EP2 478) (Atkin 2023); Khan and Dutta (2024) and Nishiura et al. (2023) (BZ target/spiral waves; mutual annihilation as the *generic* property of excitable media); Davidsen and Kapral (2003) (defect-mediated turbulence — a short, finite correlation length, **not** "no characteristic scale"); the logistic period-doubling $\leftrightarrow$ Mandelbrot real-axis correspondence and Feigenbaum universality (Feigenbaum 1978), and the Grebogi–Ott–Yorke crises (boundary / interior / attractor-merging) (Grebogi et al. 1983) used in the companion metaphor map.

**Primary reached, detail from abstract/landing (V2).** Bousquet and Slotine (2015) ($\varepsilon_c$; Remark 5 — the exact inequality is flagged for re-confirmation against the source typesetting and is *not* relied on); Koestler (1969); Varzi (SEP *Mereology*) (Varzi 2019); Kim (IEP *Causal Exclusion*; the Causal Inheritance Principle anchored to Kim (1992)); Friston (2019); Bruineberg et al. 2022; Pattee (*Physics of Symbols*) (Pattee 2001); Barrett (2017); Chalmers (2012); Bishop (2002); Sprevak (2018); Piccinini (2015); Rescorla (2013); Weinberger (2020); Ladyman and Ross (2007) (via NDPR review; primary book owed); Godfrey-Smith (2009) *Darwinian Populations* (via NDPR review); Heins and Da Costa (2022); Beck and Ramstead (2025); Maturana & Varela (demarcation) (Maturana and Varela 1980); Smeenk & Ellis (SEP *Philosophy of Cosmology*) (Smeenk and Ellis 2017); Wheeler–DeWitt (SEP *Quantum Gravity*) (Weinstein and Rickles 2024); Yao (2023); Boulding (1956) (quote + venue); McCulloch (1945) (abstract only — body paywalled, quoted from the abstract); Foerster (1981) (*objects as tokens for eigen-behaviors* — the originating eigenform thesis, confirmed via Kauffman's first-hand reportage, primary

403-gated); the Belousov–Zhabotinsky reaction and Prigogine's dissipative structures (textbook / encyclopaedia) (Nicolis and Prigogine 1977); metonymy (the part-for-whole figure is, precisely, *synecdoche* — a species of metonymy; section 6.4 uses metonymy-by-contiguity, which the handle case fits).

**Leads, not load-bearing (V3 / owed).** Putnam (1988) (primary); Godfrey-Smith 2009 *Triviality Arguments Against Functionalism* (Phil. Studies); Shagrir 2018 (semantic view); Raja et al. 2021 (*the Markov blanket trick*); the candle-flame Free-Energy exclusion case; Uzquiano's own cardinality primary; Hofstadter 1979; Koestler 1967; the exact closed form of $\varepsilon_c$; applied category theory (Fong–Spivak — the structural-register language of section 8.1); and the speculative-cosmology pointers (eternal inflation, conformal cyclic cosmology) — named from knowledge, not relied on. None of these carries weight in the argument; each is flagged where it appears.

# C Disclosure of LLM-assisted authorship

This paper was written with large-language-model assistance — drafting, literature sweeps, and adversarial verification of citations — under the author's direction and verification. Consistent with current venue norms (LLMs may be tools but not authors, and non-trivial AI use is disclosed), the LLMs are tools, not authors; the author is solely responsible for every claim. The verification apparatus that makes this disclosure safe is itself part of the method: the verification levels of section B were assigned by first-hand checks, and at least one machine-produced research conclusion was caught over-reaching and excluded on review (recorded in the companion claim ledger, §F).

# D A worked near-decomposability verdict on a wave medium

To make the partition-relativity of section 2.2 concrete — and to exhibit one *fully-worked, pre-registered* verdict (the gap section 8.2 flags) — the demarcation was run on a reaction–diffusion "wave" medium: the 2D complex Ginzburg–Landau equation,

$$\partial_t A = A + (1 + ic_1)\nabla^2 A - (1 + ic_2)|A|^2 A,$$

on a $128^2$ periodic grid.

**Pre-registered before measuring.** *Partition:* $4 \times 4$ spatial blocks (block size $B = 32$ cells). *Signature:* the spatial correlation length $\xi$ (radial autocorrelation of $\mathrm{Re}\, A$, $1/e$ drop). *Verdict rule:* near-decomposable into the blocks iff $\xi \ll B$ (blocks quasi-autonomous, inter-block coupling weak); not, if $\xi \gtrsim B$ (long-range coupling). *Two regimes:* Benjamin–Feir-stable ($1 + c_1c_2 > 0$; here $+1.25$) and defect-turbulent ($1 + c_1c_2 < 0$; here $-3.0$).

**Measured (three seeds each).** Stable regime: $\langle|A|\rangle \approx 0.99$ (coherent plane wave), $\xi/B \approx 0.80$ (0.74–0.85) $\rightarrow$ *not* block-decomposable. Defect-turbulent regime: $\langle|A|\rangle \approx 0.61$ (fluctuating), $\xi/B \approx 0.12 \rightarrow$ *near-decomposable*. The $\sim 7\times$ separation is robust across seeds.

**What it shows — two things, and no more.** (1) It *corrects* an over-quick claim in an earlier draft of section 4.2: under a spatial-block partition the *coherent* regime is the one that resists decomposition (long-range order), and the *turbulent* one decomposes — the reverse of "organised = decomposable." (2) It is a *measured* instance of the central claim (section 2.2): the near-decomposability verdict is a property of the *(system, regime, partition, signature)* tuple, not of the system — so much so that it inverts the intuitive reading. Its weight, as with section E, is small: a single toy model, one signature, three seeds — it demonstrates the *method* and its partition-relativity, and makes no claim about turbulence, order, or the universe.

It is a worked *verdict*, not a worked *exclusion* (the coherent field is merely not-decomposable *under this partition*, not under every admissible one), so the section 8.2 request for a confirmed all-partitions exclusion remains only partly met.

# E A worked self-falsification in a built holonic system

The discipline of section 3 is not only prescriptive: I have applied it, with a null result, to a holonic system I built. The episode is small and the result negative — which is exactly why it is worth reporting, because it shows the method enacted on my own commitments rather than only urged on others.

**The system and its falsifiable teeth.** Phyriad is a holonic computational substrate — a containment tree of holons (section 5.2) over a lateral stigmergic field. To give the holarchy *dynamical* content a flat model would lack, I modelled its node dynamics as a coupled-quadratic tree iteration,

$$z_{n+1}(H) = \alpha \cdot z(H)^2 + \beta(H) \cdot \sum_c w(c) \cdot z(c) + \delta(H) \cdot z(\text{parent}) + \omega(\text{task}, \text{field}), \quad \text{with } z = \tau + i\gamma,$$

where $\tau$ is latency and $\gamma$ a pheromone gradient. This was declared up front to be a special case of an established equi-M / network-Mandelbrot iteration — *not* new mathematics. The load-bearing term is the **quadratic self-interaction** $\alpha \cdot z^2$: the holonic-specific nonlinearity, the part of a holon's dynamics that is *more* than a coupled sum of its neighbours and parent. If holonic organisation adds dynamical content here, $\alpha$ must come out measurably nonzero.

**The program, and the null.** The null was established in four runs of escalating stringency, all anchored in the repository and reproducible.[2]

1. **Overload calibration.** The first pass (75 recorded traces, an LLM collaborator, 2026-06-10) fitted $\alpha \approx +0.000$ in all 75 — but it never *forced* a role: the role-emergence rule silently recollapsed interior nodes to the leaf default, so no role was actually held fixed. A rebuilt harness forces the role (bypassing the emergence rule) and drives genuine overload ($\tau \in \{1.5, 2.0, 3.0\}$, measured occupancy sustained above 1.2). $\alpha \approx 0$ in every cell; the linear fit holds at $R^2 \approx 0.999$ and attributes the dynamics to the linear coupling term ($\beta \approx 0.95$).

2. **Positive control (a synthetic "plastic" coin).** A trace generated from the fitter's *own* equation with a planted $\alpha = 0.3$ is recovered to $\lesssim 10^{-11}$. The instrument is not blind to a nonzero quadratic.

3. **Injected "metal" coin.** A genuine quadratic $\alpha\tau^2$ was injected into the *real measured latency* (extra busy-work each tick, timed by the real clock), *not* into the fitter's arithmetic. The injection demonstrably enters the dynamics — the operating $\tau$ rises from a sub-overload base into overload, and an unstable setting ($2\alpha\tau^* > 1$) collapses the trajectory exactly as the fixed-point analysis predicts — yet the fitted $\alpha$ stays $\approx 0$: the injected nonlinearity is *absorbed into the linear coupling coefficient* $\beta$, not recovered as $\alpha$.

4. **Diagnosis: an identifiability wall.** Near a stable fixed point the observable $\tau$ is nearly constant, so the regressors ($\tau^2$, the neighbour and parent terms, the constant) are collinear;

[2] Each number below is a *claim until re-run.* Tooling under `tools/phyriad_set_lab/`; raw outputs under `pslab_run/`. Build `holarchy_trace`; overload calibration `--force-role <role> --target-tau {1.5,2.0,3.0}` then `fit_coefficients.py`; plastic control `positive_control.py`; metal injection `--inject-alpha {0.0,0.1,0.3,0.5}`.

real measurement noise then places the quadratic's weight on the linear terms. The null is therefore not only "$\alpha \approx 0$" but "$\alpha$ *unidentifiable in-regime*."

**Caveat — the coin is *partial*, not clean metal.** The injection shares the *quadratic form* the fitter assumes (one cannot test recovery of a quadratic without injecting one), and it enters $\tau$ (the real part) while the fitter models the complex $z^2 = \tau^2 - \gamma^2 + 2i\tau\gamma$, so even a clean recovery would map the injected coefficient to $\alpha$ only up to a $\gamma^2$ term. The *mechanism* is real (work, clock, $\tau$); the *recovery* is regime-limited.

**What it shows, and what it does not.** This is a small, negative, single-system result, and I claim no more for it. It is *not* evidence *for* the holonic thesis; if anything it is a mild disconfirmation of holonic-specific nonlinearity *in this system's operating regime*. Its value is methodological, and narrowly so: it exercises the *pre-registration discipline* of section 3 — models fixed in advance, measured, the null reported rather than buried, with a positive control to show the instrument can see the signal when it is planted — but it tests a specific holonic-nonlinearity sub-claim (is $\alpha$ nonzero *and recoverable*?), *not* the full near-decomposability demarcation (no partition was placed in or out of the contraposition class). The identifiability wall is, moreover, a *measured instance* of the underdetermination section 3.4 declares open: the holonic-specific term is not merely zero but *behaviourally indistinguishable from its linear absorption* — the empirical face of the triviality the paper declines to claim solved. It exhibits that open problem; it does not resolve it. It is a demonstration of the pre-registration habit that keeps a universal-sounding framework off the General Systems Theory side of Popper's line (section 3.5) — not a worked instance of the demarcation criterion itself, which this paper does not yet exhibit on a real system (section 8.2). A framework that can self-administer a null, and say so, is one whose positive claims are worth more.

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